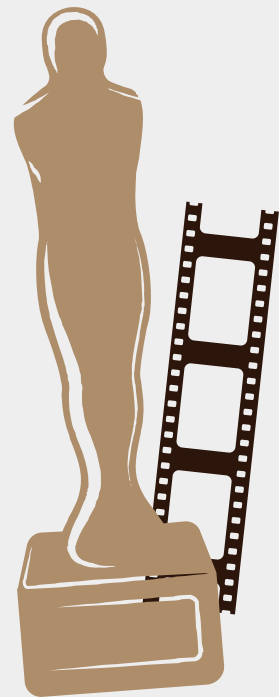


IMDB DATA ANALYSIS

Jasmine Jordan-Lake & Charlotte Matthews



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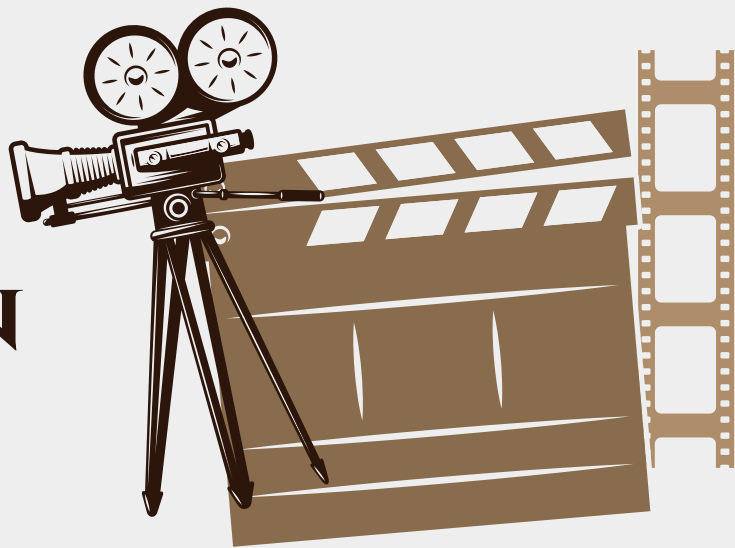
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01

INTRODUCTION

Purpose, scope, and goals



★ INTRODUCTION ★



“IMDb (Internet Movie Database) is the world's most popular and authoritative source for movie, TV and celebrity content. Find ratings and reviews for the newest movie and TV shows.”

- IMDb

IMDb

- To determine a movie's rating, IMDb users cast votes for a movie which are then summarized into a single rating of the movie out of ten.
- For one movie on IMDb, we can get loads of information about it like it's IMDb rating, box office earnings, runtime, classification, actors, directors, etc.



What to watch

| Top picks >

TV shows and movies just for you

[Sign In](#)

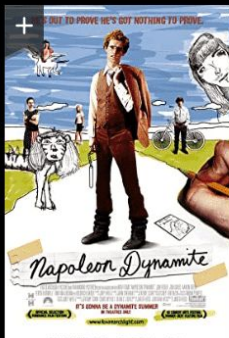


★ 7.5 ☆

American Gigolo

[Watch options](#)

▶ Trailer

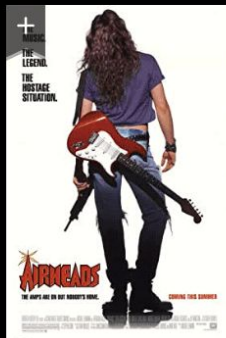


★ 6.9 ☆

Napoleon Dynamite

[Watch options](#)

▶ Trailer

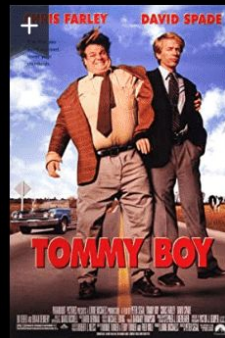


★ 6.1 ☆

Airheads

[Watch options](#)

▶ Trailer



★ 7.1 ☆

Tommy Boy

[Watch options](#)

▶ Trailer



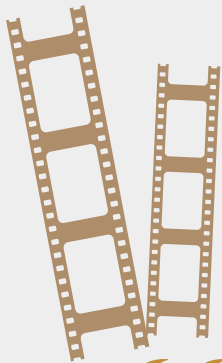
★ 6.6 ☆

Half Baked

[Watch options](#)

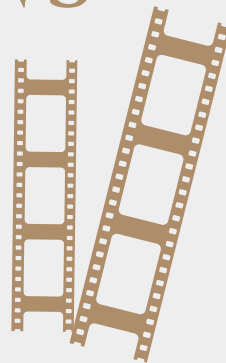
▶ Trailer





669,946 MOVIES AND SHOWS

have been rated on IMDb.



- During this analysis, we utilized an IMDb Dataset from data.world to examine and visualize patterns, correlations, and potential trends.
- This dataset consists of 5273 observations and 27 features.



KEY QUESTION:

How have audience preferences and film industry norms changed over time?

What impacts have these shifts had, notably on IMDb ratings and Box Office earnings?

ADDITIONAL QUESTIONS TO EXPLORE:

- ❖ How has the IMDb rating and runtime of movies **changed over time** ? Does IMDb tend to **favor certain eras** of movies more than others?
- ❖ How are the IMDb scores **distributed** throughout the dataset? What are the **trends/patterns** in the audience ratings?
- ❖ Is there a **correlation** between Box Office earnings and IMDb ratings? If so, what is the relationship between critical reception and financial success?
- ❖ What are the runtime and IMDb **rating differences** between different intended audience ratings?
- ❖ Are there **correlations** between runtime and IMDb ratings? How does the **length of a film impact** the audience rating of the film?

★ DATA ANALYSIS: OBJECTIVES ★

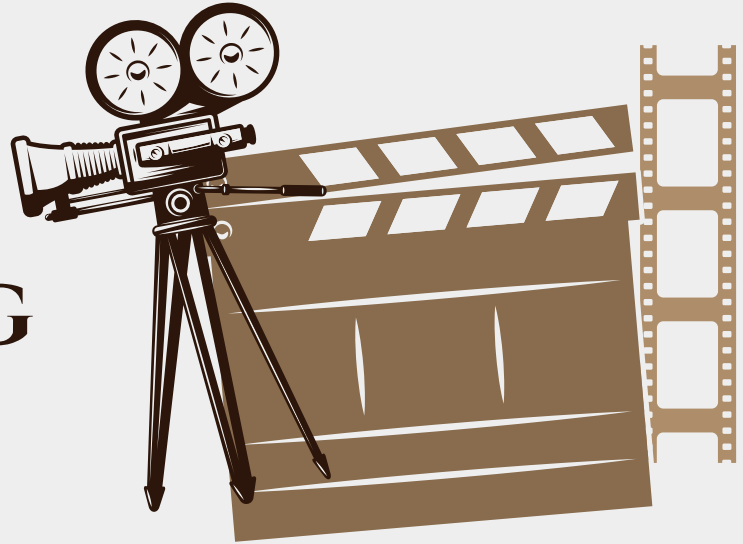
- ★ Obj. 1: Investigate IMDb Rating Trends Over Time and Analyze Average Runtime.
- ★ Obj. 2: Analyze IMDb Rating Frequency and Distribution of IMDb Movies Across the Dataset.
- ★ Obj. 3: Investigate the Correlation Between IMDb Rating vs. Box Office.
- ★ Obj. 4: Look at the relationship between film runtime and IMDB ratings to uncover patterns or trends between film duration and audience perception.
- ★ Obj. 5: Analyze the variations in IMDB ratings and runtime defined by a film's rating.



02

DATA CLEANING

Cleaning and processing of the raw data



LOOK AT RAW DATASET

27

Features

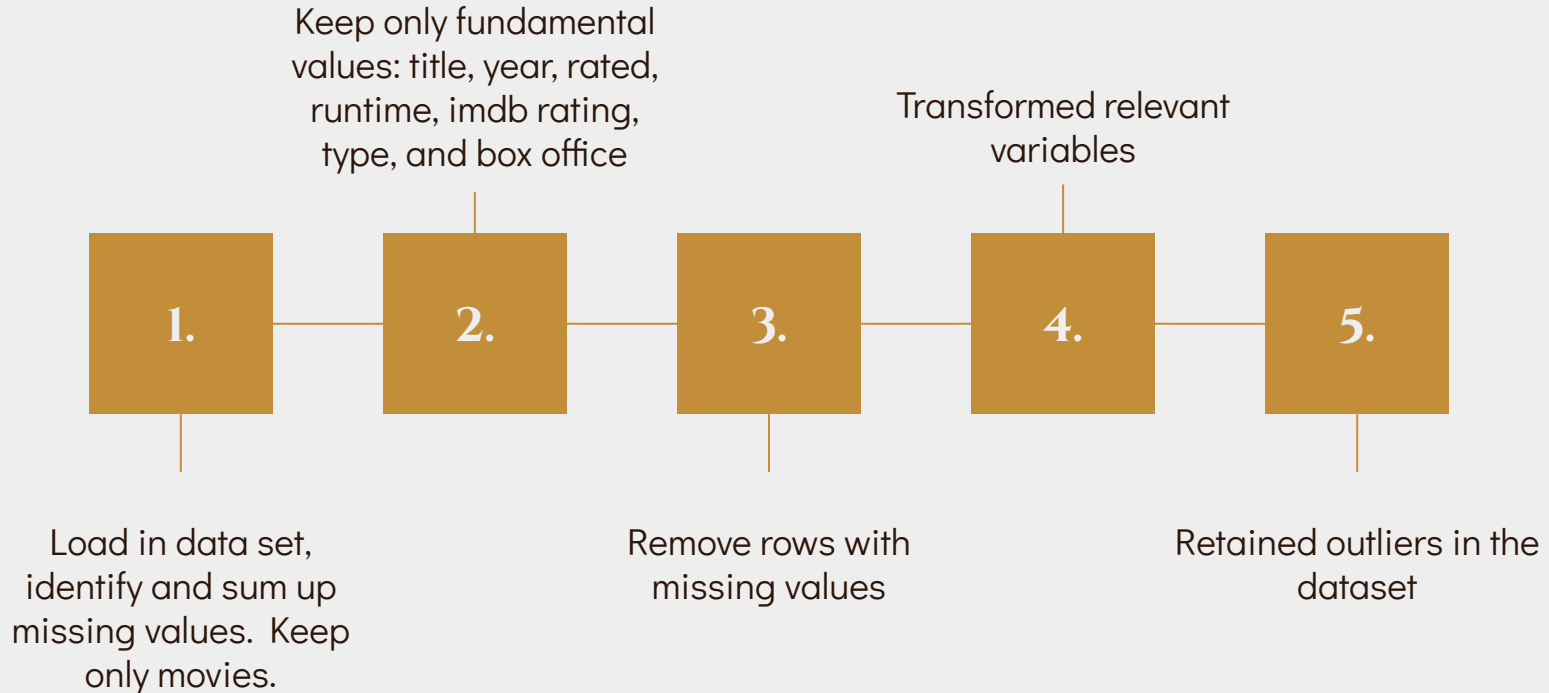
5273

observations

★ THERE ARE 27 FEATURES (COLUMNS) IN OUR DATASET: ★

- | | |
|--------------|---------------------------|
| 1. Title | 14. Poster |
| 2. Year | 15. Ratings.Source (IMDb) |
| 3. Rated | 16. Ratings.Value (/10) |
| 4. Released | 17. Metascore |
| 5. Runtime | 18. imdbRating |
| 6. Genre | 19. imdbVotes |
| 7. Director | 20. imdbID |
| 8. Writer | 21. Type |
| 9. Actors | 22. DVD |
| 10. Plot | 23. BoxOffice |
| 11. Language | 24. Production |
| 12. Country | 25. Website |
| 13. Awards | 26. Response |
| | 27. tomatoURL |

★ WHAT NEEDS TO BE DONE ★



POST-CLEANING

7

Features

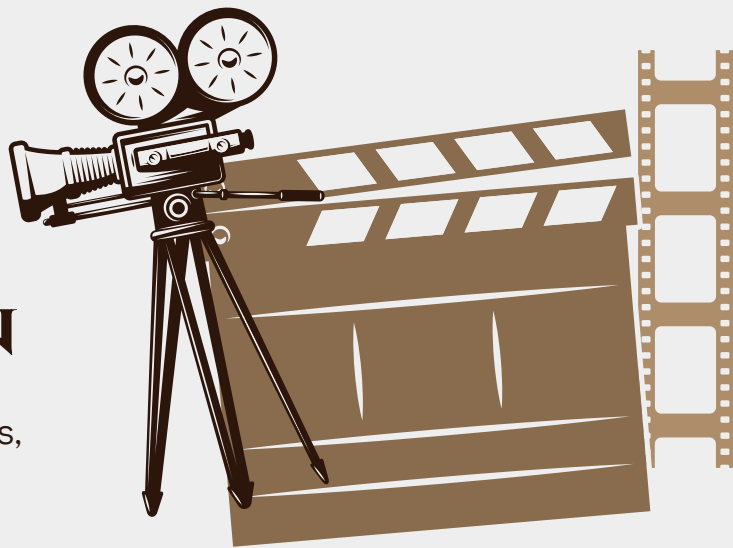
2550

observations

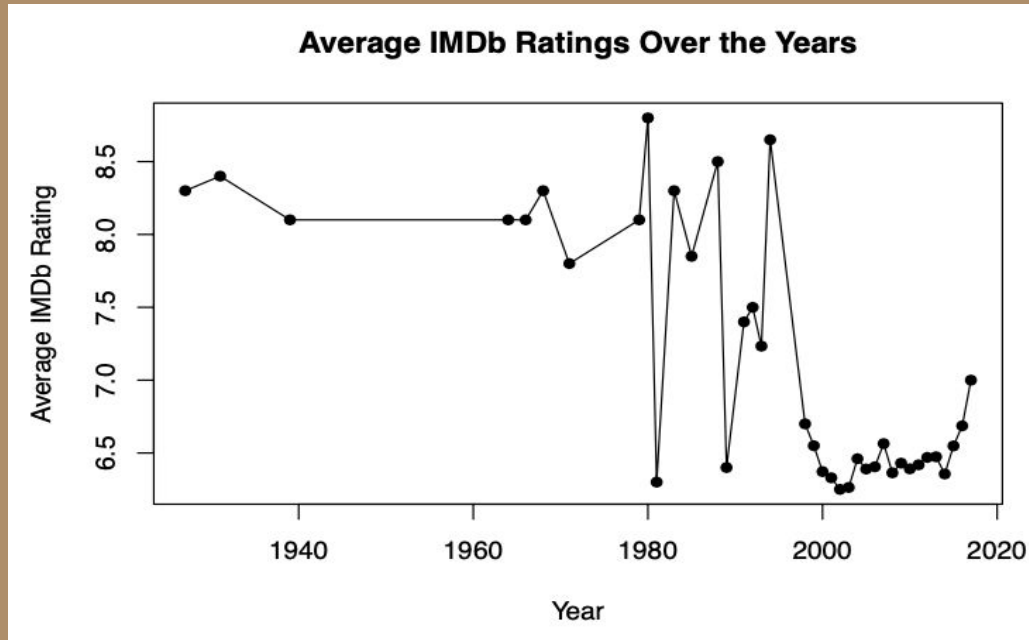
03

DATA VISUALIZATION

Visual representations to study patterns, trends,
and relationships

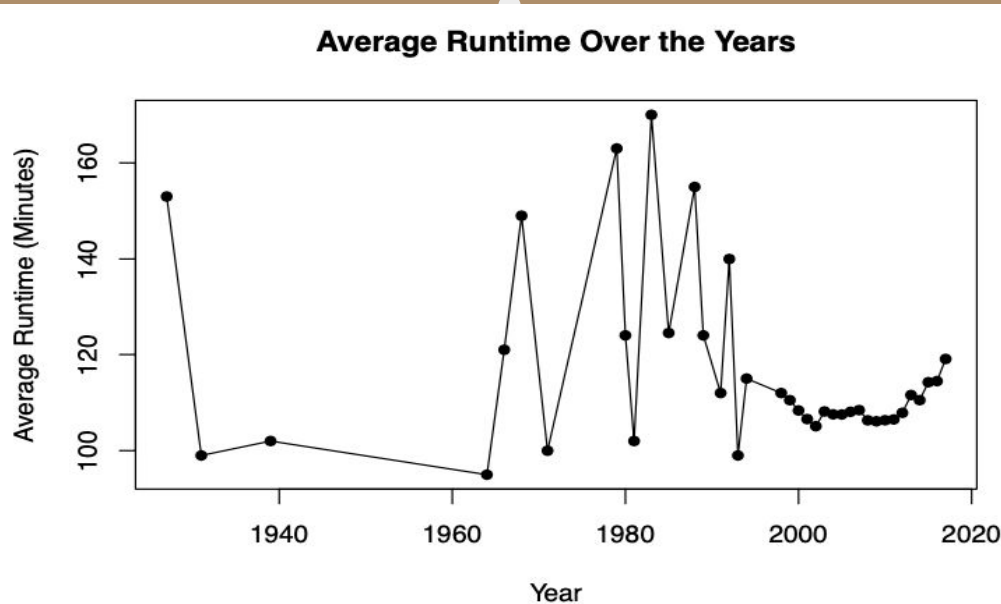


★ AVERAGE IMDB RATINGS OVER THE YEARS ★



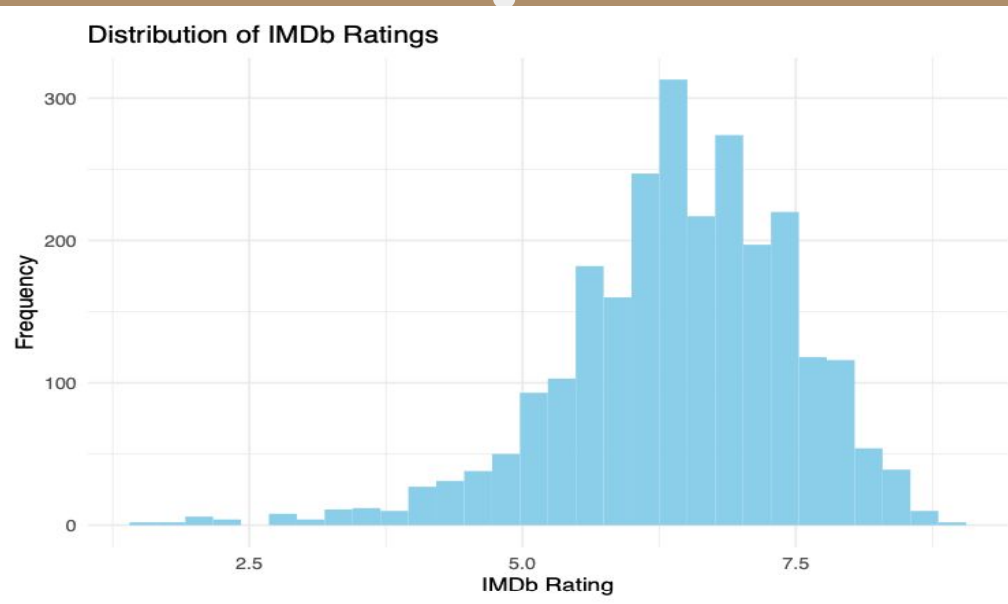
- **1920s-1970s:** Average IMDb ratings are significantly **high**
 - “Cinematic Golden Age”
- **1980s-2000s :** High **variability**
- **2000s-Current:** **Convergence** , lower ratings
- **Homogenization** of movie preferences and the evolution of movie dynamics

★ AVERAGE RUNTIME OVER THE YEARS ★



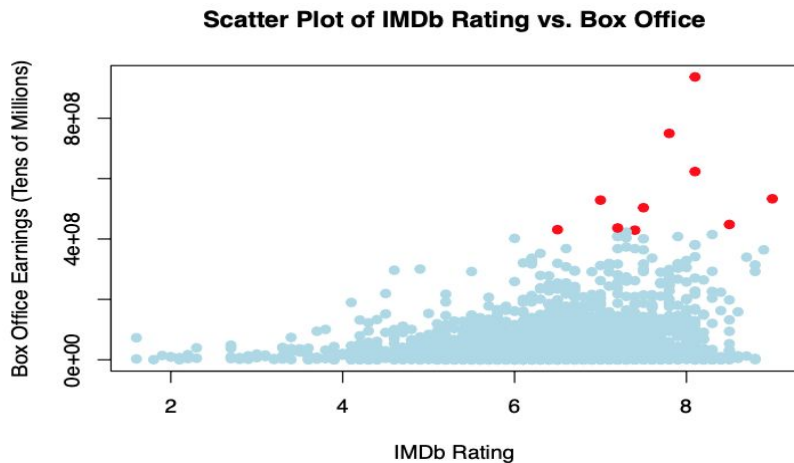
- **Pre2000s** : Many shifts in the runtime for movies
 - **evolving narrative styles** and audience expectations
- **2000s-Current**: **Convergence** , lower runtimes
 - Patterns of **modern day viewing habits** with growing digital platforms and easier access to movies
 - Demand for more concise movies and storylines establishing a standard for runtime

★ DISTRIBUTION OF IMDB RATINGS ★



- Majority of the movies have an above average rating spanning between ~5.8 to ~7.3
- Outliers exist outside of this range
- The median rating is around 6.6

★ IMDB RATING VS. BOX OFFICE ★

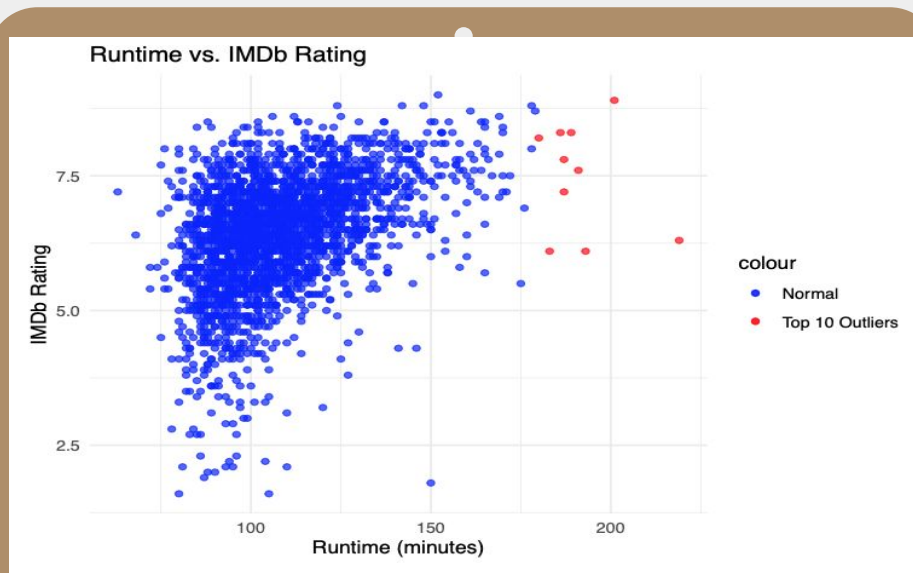


- A small positive relationship
- In red we marked the 10 largest outliers
- Exceptionally high box office earnings, but their IMDb ratings aren't necessarily the highest.

Example outlier:

- Star Wars: Episode I - The Phantom Menace
 - Rating: 6.5
 - BO: 431 mil

★ RUNTIME VS. IMDB RATING ★

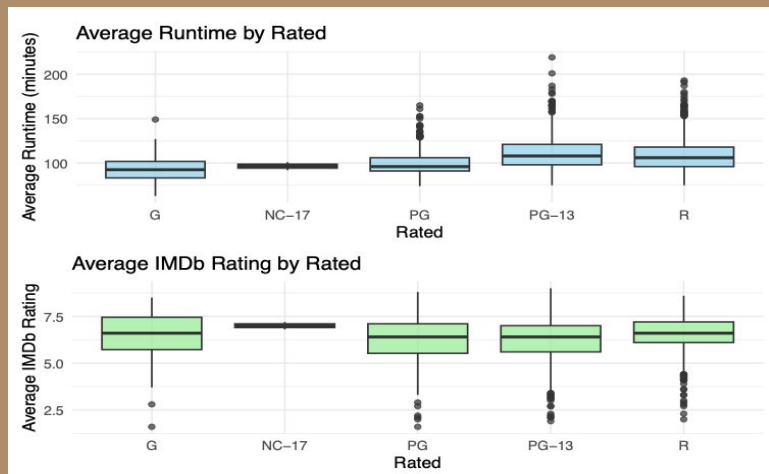


- Fairly positive correlation, longer runtimes typically receiving higher IMDb ratings
- Outliers based on the farthest distance from the median runtime and IMDb rating exhibit notable exceptions.

Example outliers:

- "Gods and Generals" with a lengthy runtime of 219 minutes garnered a modest IMDb rating of 6.3
- "The Wolf of Wall Street," with a shorter runtime compared to "Gods and Generals," received a significantly higher IMDb rating of 8.2.

★ AVERAGE RUNTIME/RATING BY RATED ★

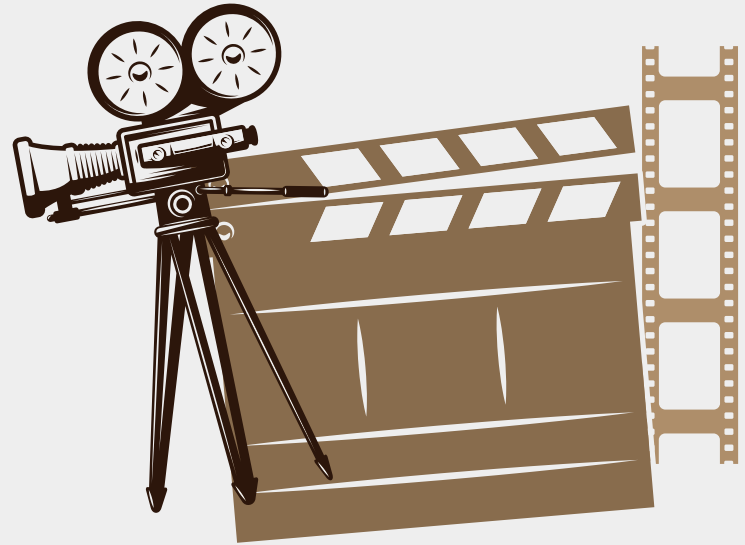


- Not Included: “UNRATED AND NOT RATED”
- **G and PG** films **runtimes** are **generally lower**, reflecting their suitability for family audiences.
- **PG-13 and R-rated** films tend to have slightly **longer** average runtimes, indicative of their appeal to older audiences and exploration of more mature themes.
- NC-17 films exhibit minimal variation.
- **G and R** rated films have the **slightly-higher edge** over the PG and PG-13 ones (with NC-17 again having a low representation).
- **Diverse tastes and preferences** of viewers

04

ACTIONABLE INSIGHTS AND LIMITATIONS

Limitations, implications for future actions



ACTIONABLE INSIGHTS

Producers and filmmakers:

- Utilize analysis insights to inform **production decisions**.
- **Optimize runtime** based on audience preferences.
- **Prioritize elements** enhancing audience perception.

Marketers:

- Utilize IMDb ratings correlation with box office earnings.
- Develop **effective marketing** strategies.
- **Maximize** movie revenues and box office performance.

Analysts:

- Explore **underlying factors** of observed trends.
- Gain **deeper insights** into film industry dynamics.

LIMITATIONS

- Coverage does not fully represent the entire range of films released over the years.
- IMDb ratings are inherently subjective.
 - Influenced by various factors, including personal biases, cultural backgrounds, and viewing experiences.
- Correlation $\neq$ causation

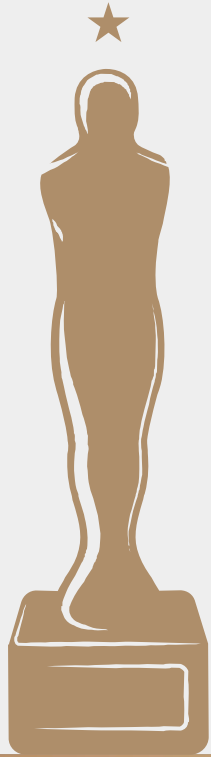


SOURCES

“IMDB Top 250 Lists and 5000 plus IMDB Records - Dataset by Studentoflife.” Data.World, 10 Feb. 2020, data.world/studentoflife/imdb-top-250-lists-and-5000-or-so-data-records.

Kosourova, Elena. “A Guide to R Regular Expressions with Examples.” DataCamp, DataCamp, 14 Oct. 2022, www.datacamp.com/tutorial/regex-r-regular-expressions-guide.

“Ratings, Reviews, and Where to Watch the Best Movies & TV Shows.” IMDb, IMDb.com, www.imdb.com/. Accessed 2 May 2024.



ANY
QUESTIONS?

